

From pest to pollinator

Using the mosquito transcriptome to investigate floral humidity as a foraging cue for insect pollinators

Massimiliano Jacob Spensley

Supervised by Dr. Heather Whitney and Prof. Davide Pisani



Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

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Mosquito-borne diseases

Mosquitoes are vectors for some of the deadliest infectious diseases worldwide...

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Aedes aegypti

Primary vector for **yellow fever**, **dengue**, **zika**, and **chikungunya** viruses.

Mosquito-borne diseases

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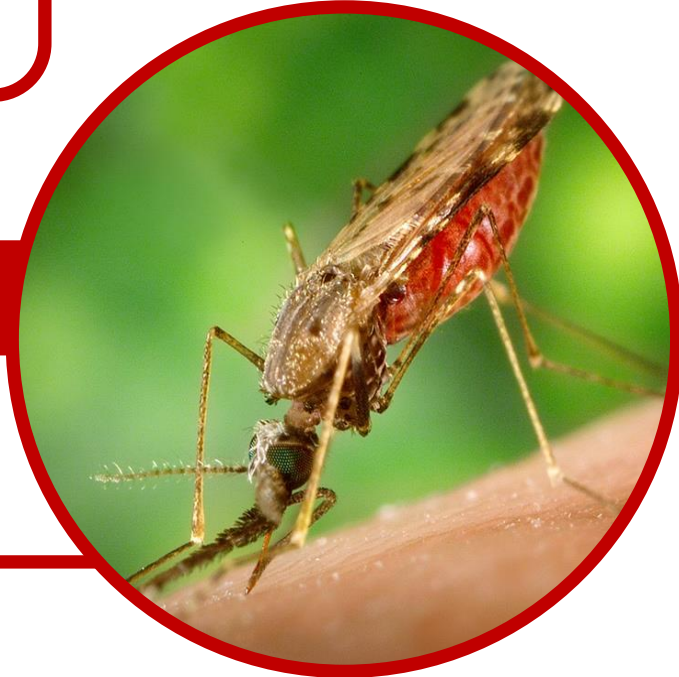


Aedes aegypti

Primary vector for **yellow fever**, **dengue**, **zika**, and **chikungunya** viruses.

Anopheles gambiae

Responsible for the majority of the global **malaria** burden.



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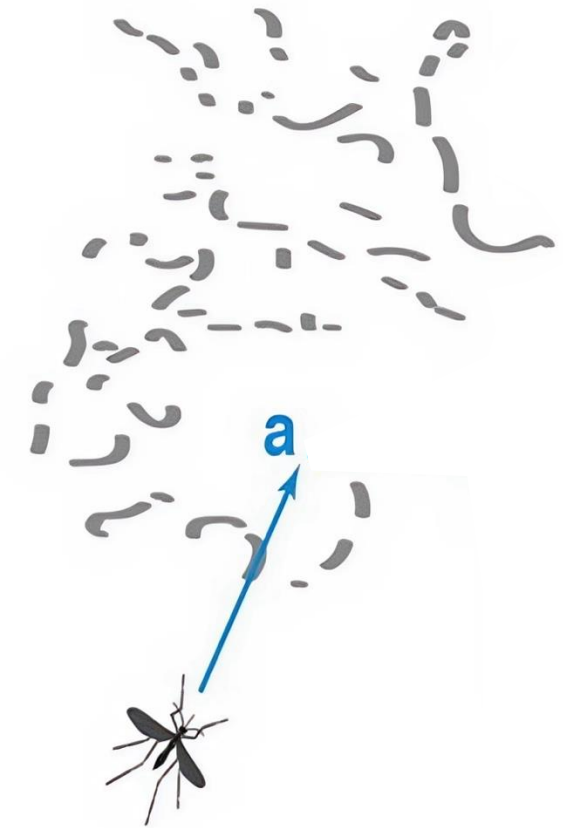
Floral humidity and insect pollinators

Mosquitoes as pollinator models

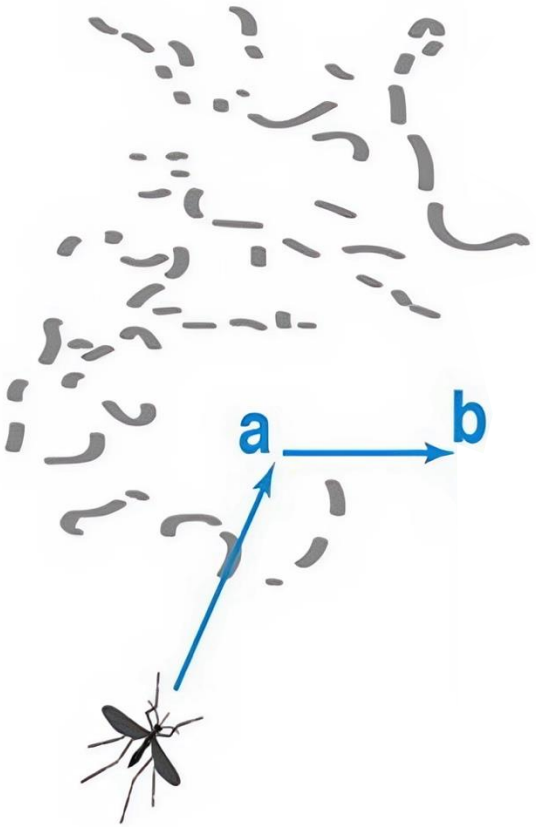
Project plan

Host location

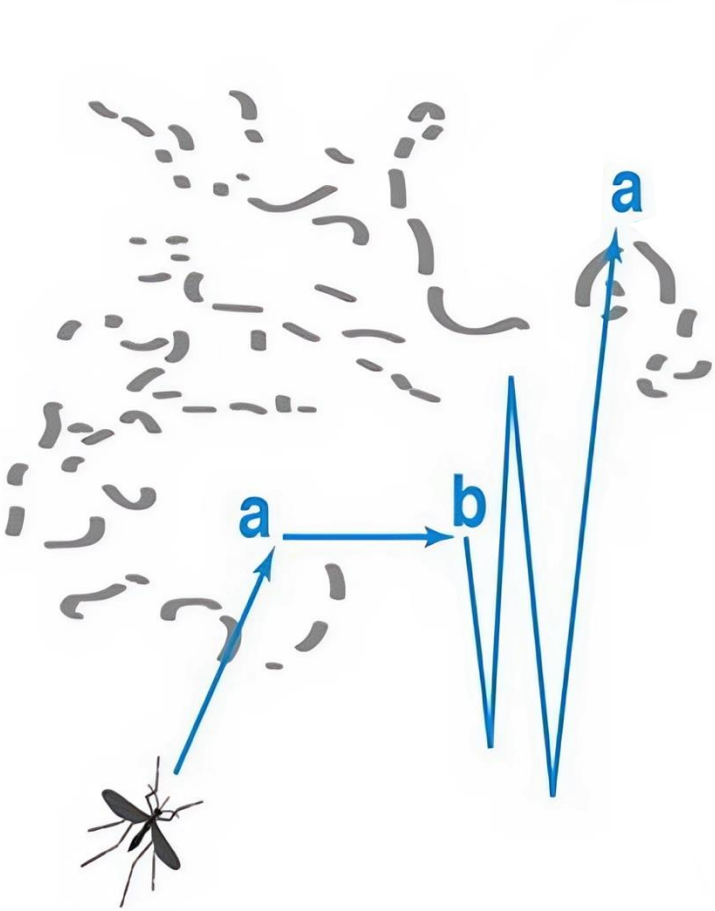
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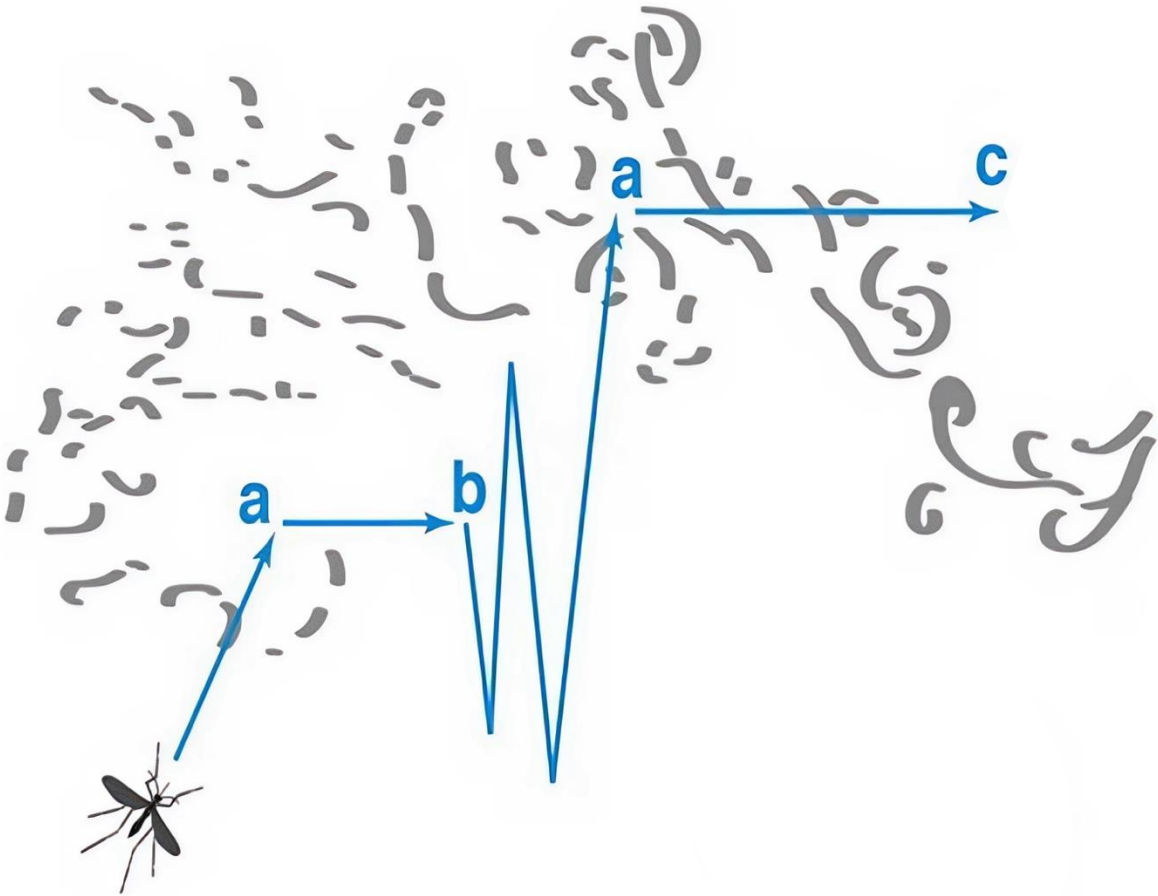
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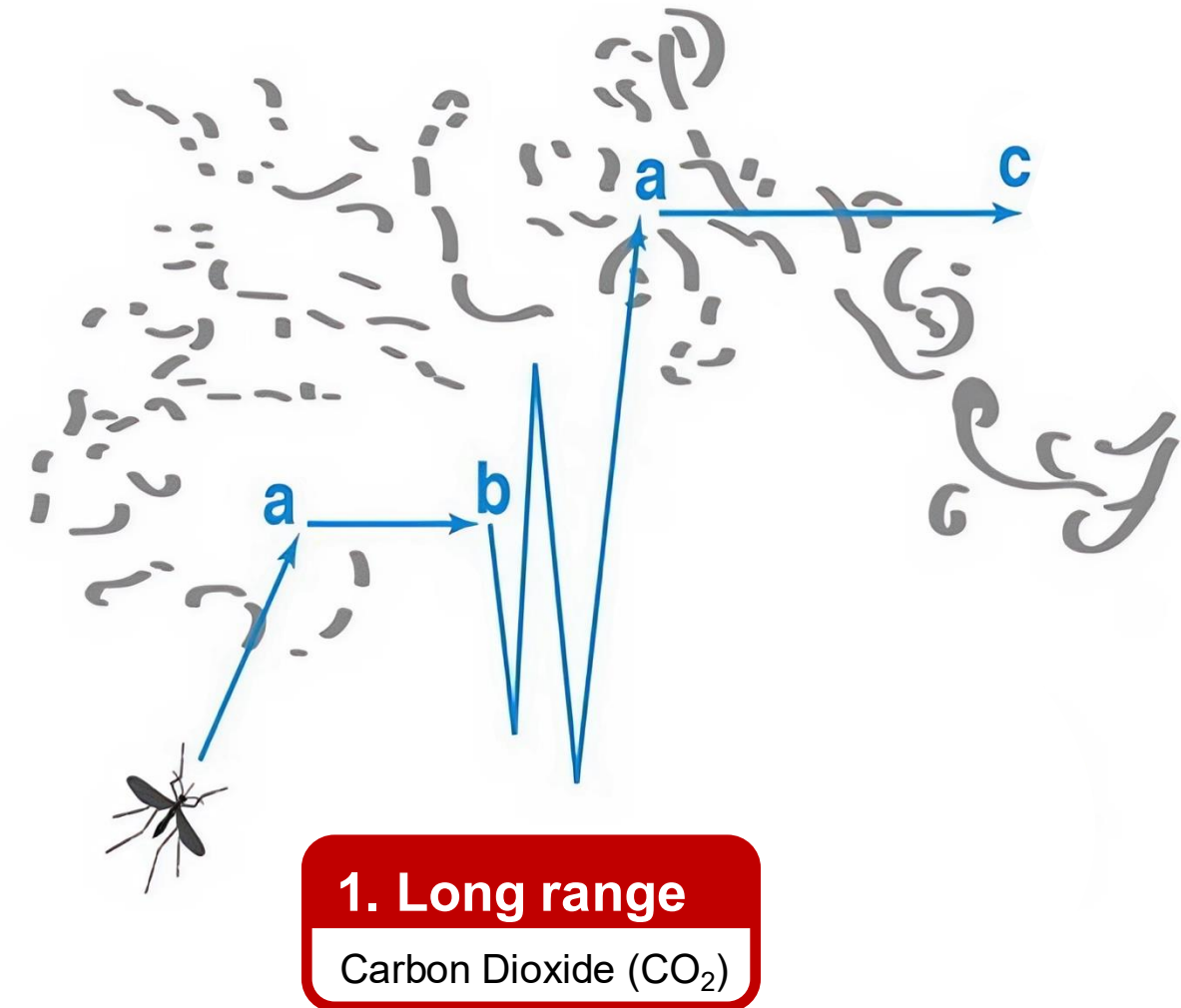
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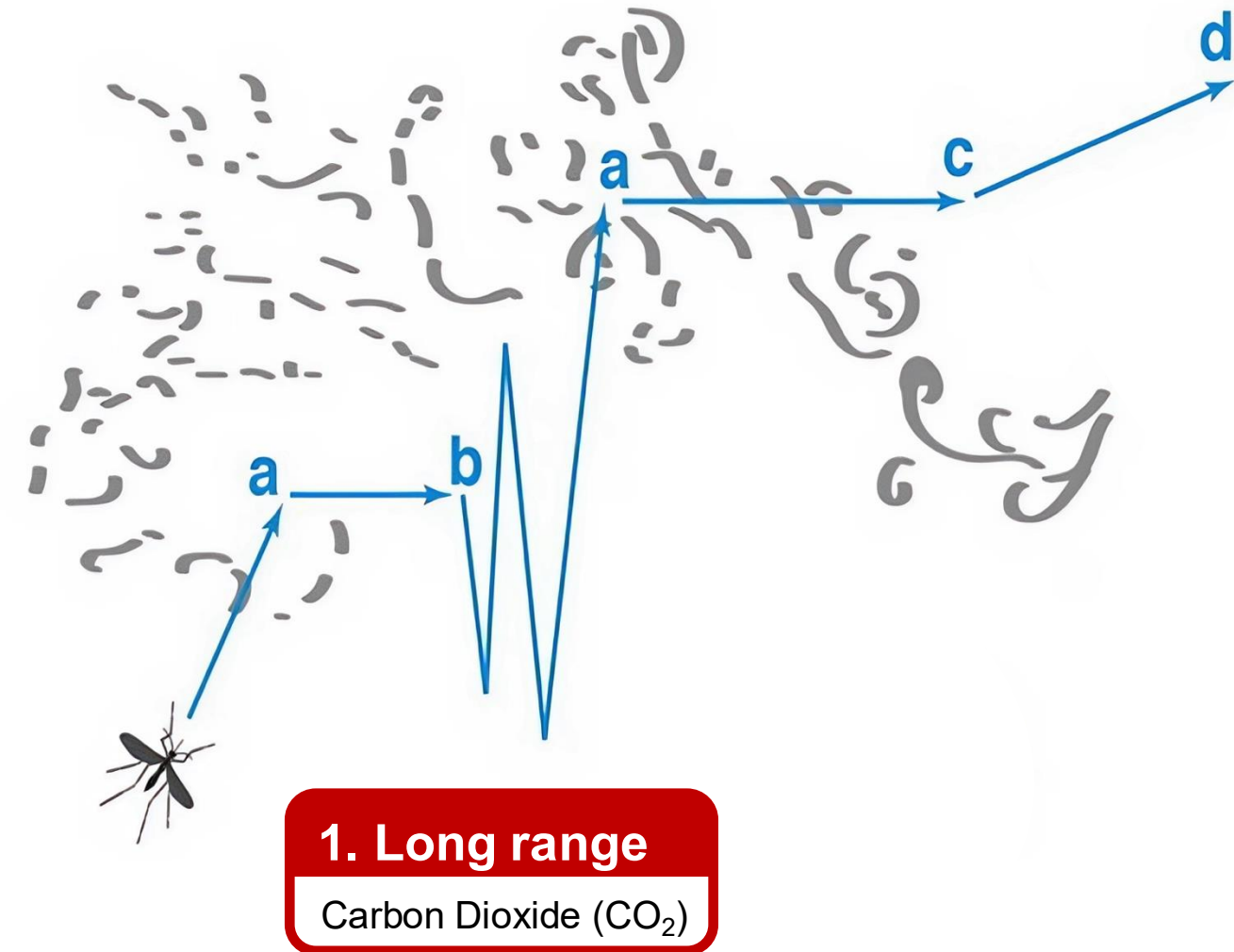
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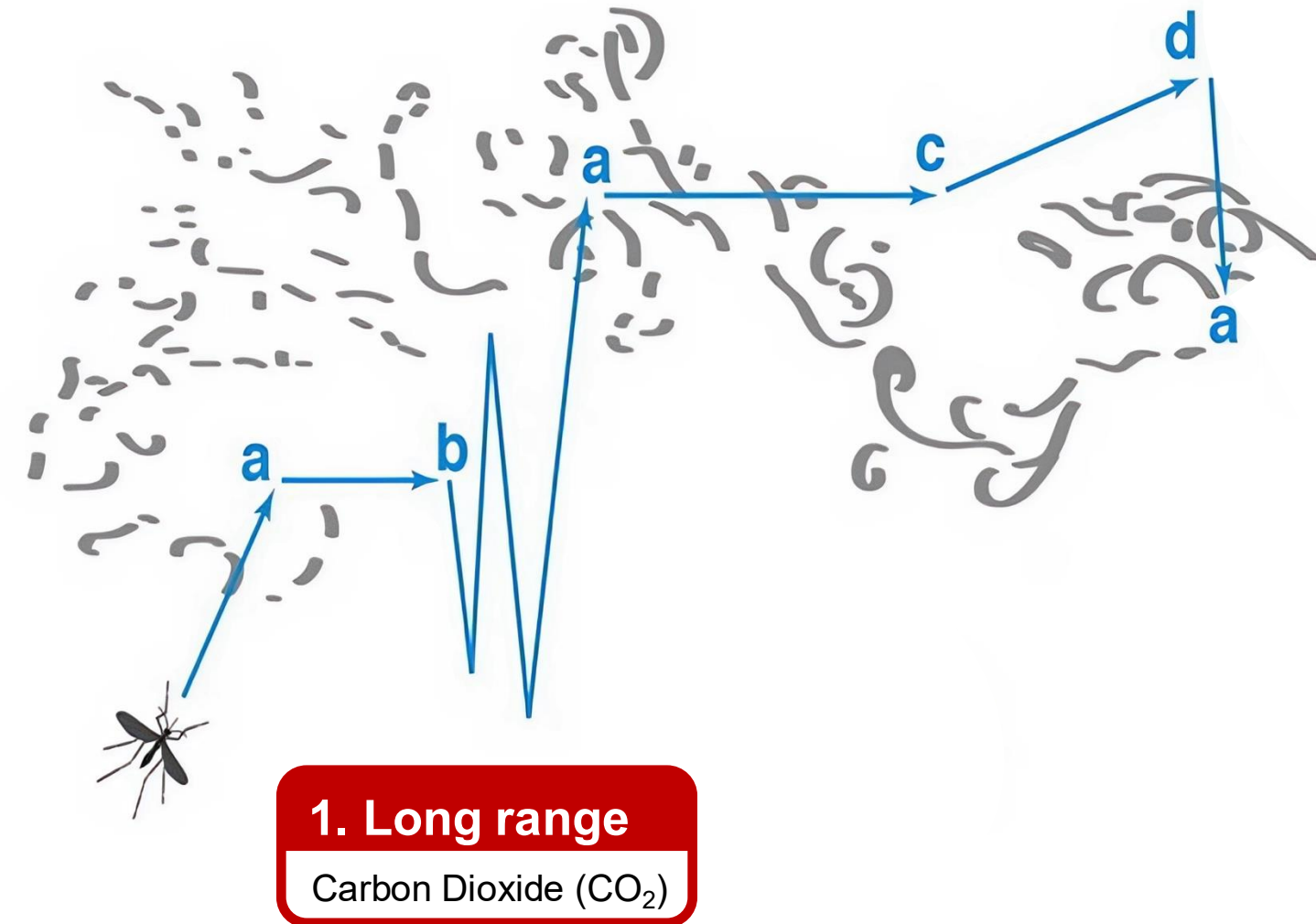
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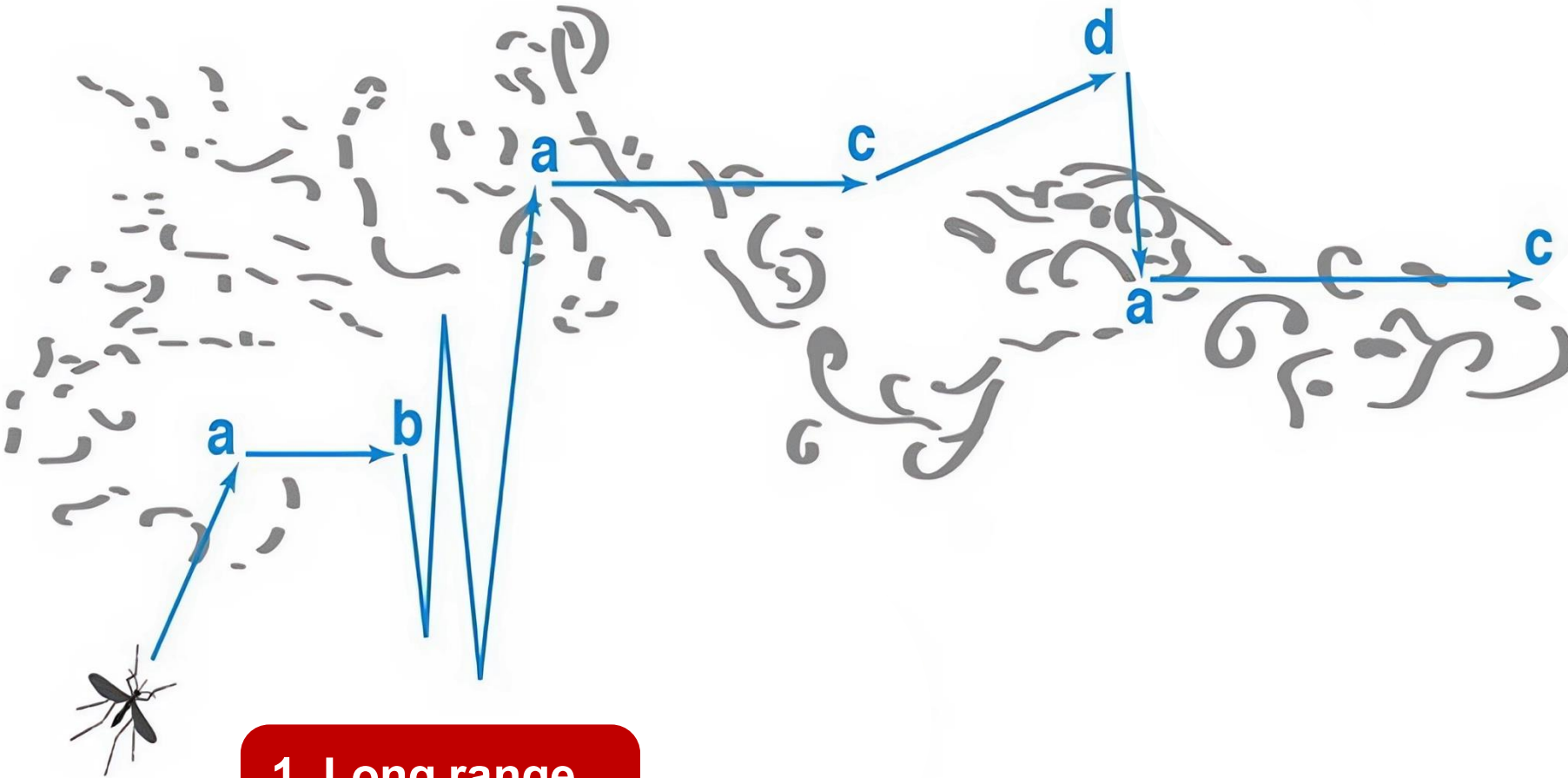
Host location



Host location



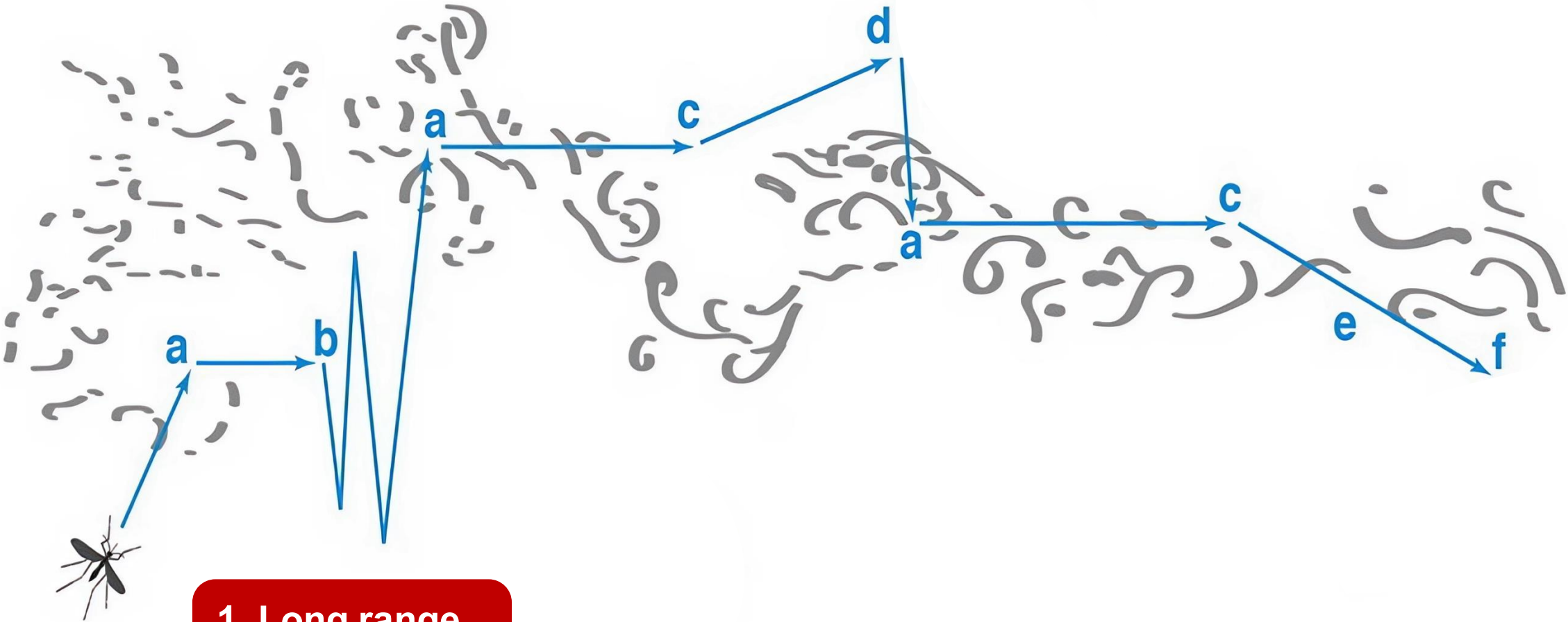
Host location



1. Long range

Carbon Dioxide (CO₂)

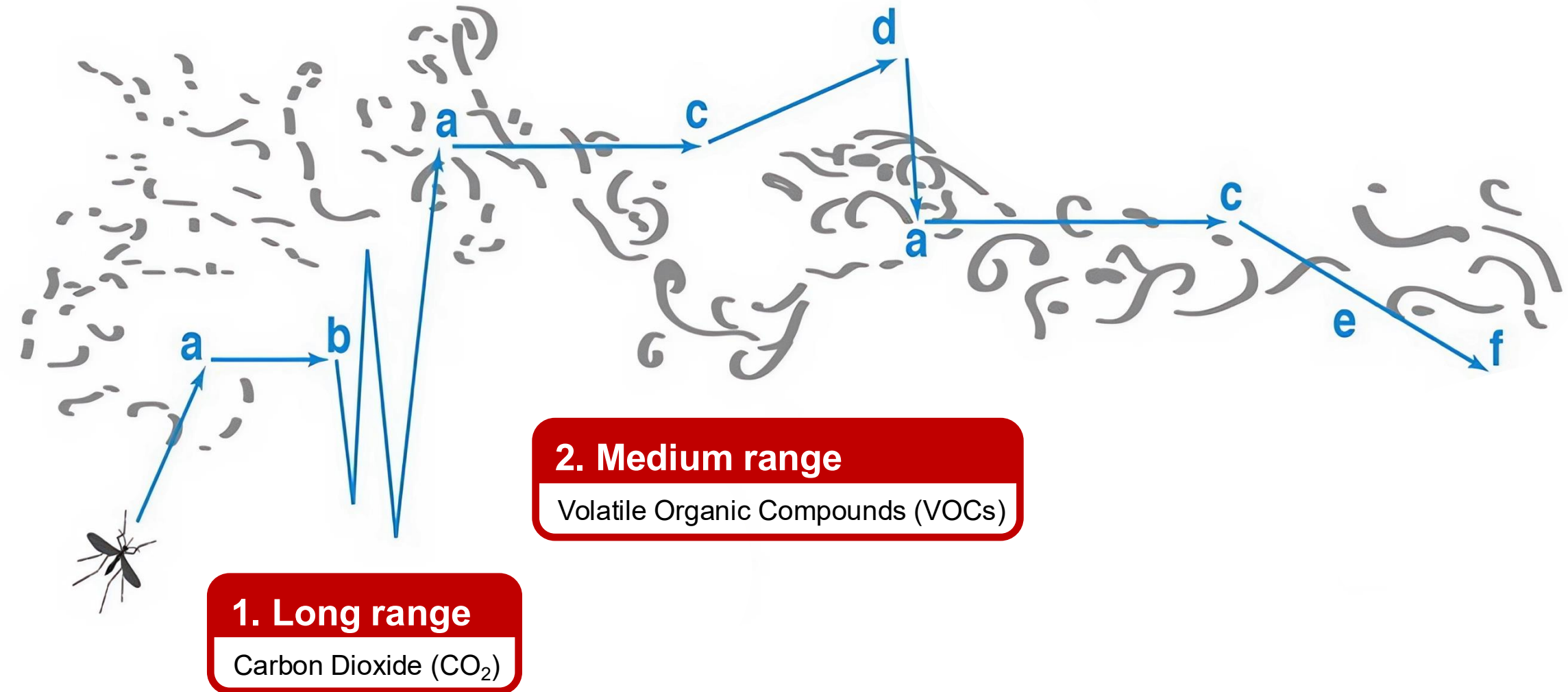
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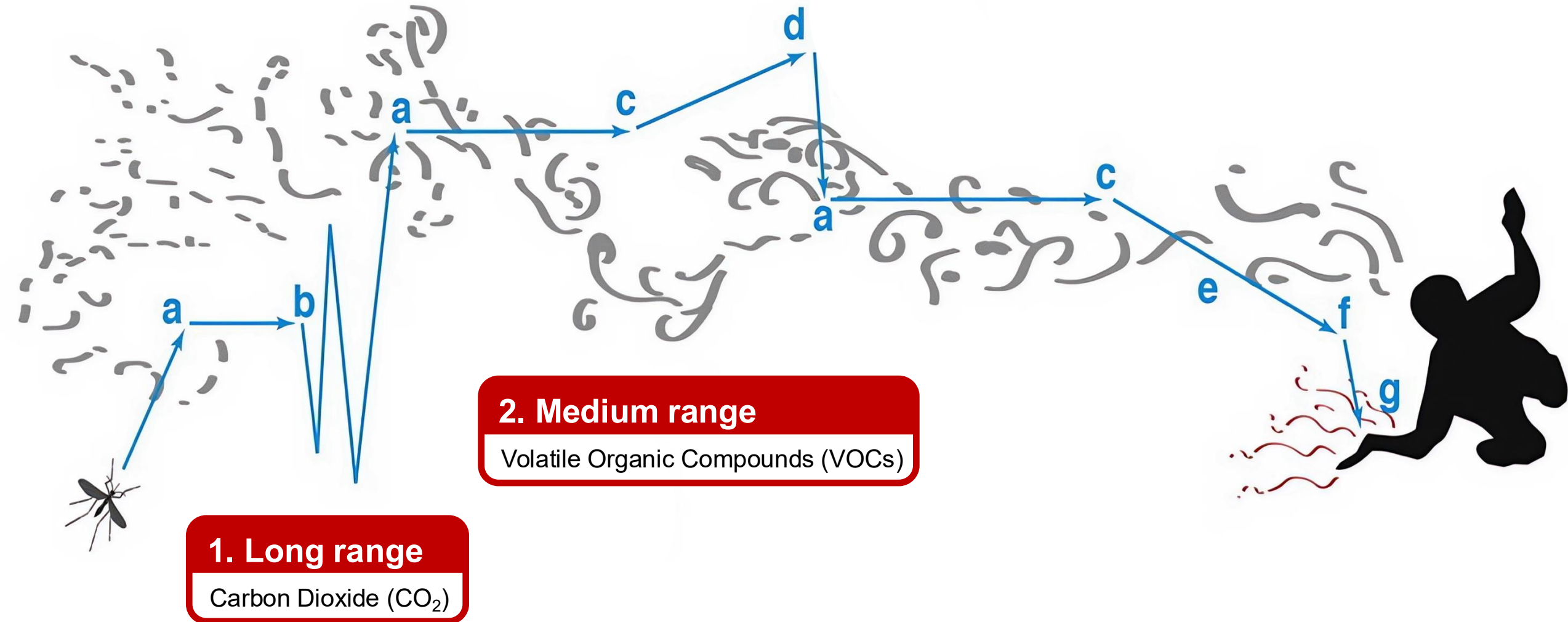
1. Long range

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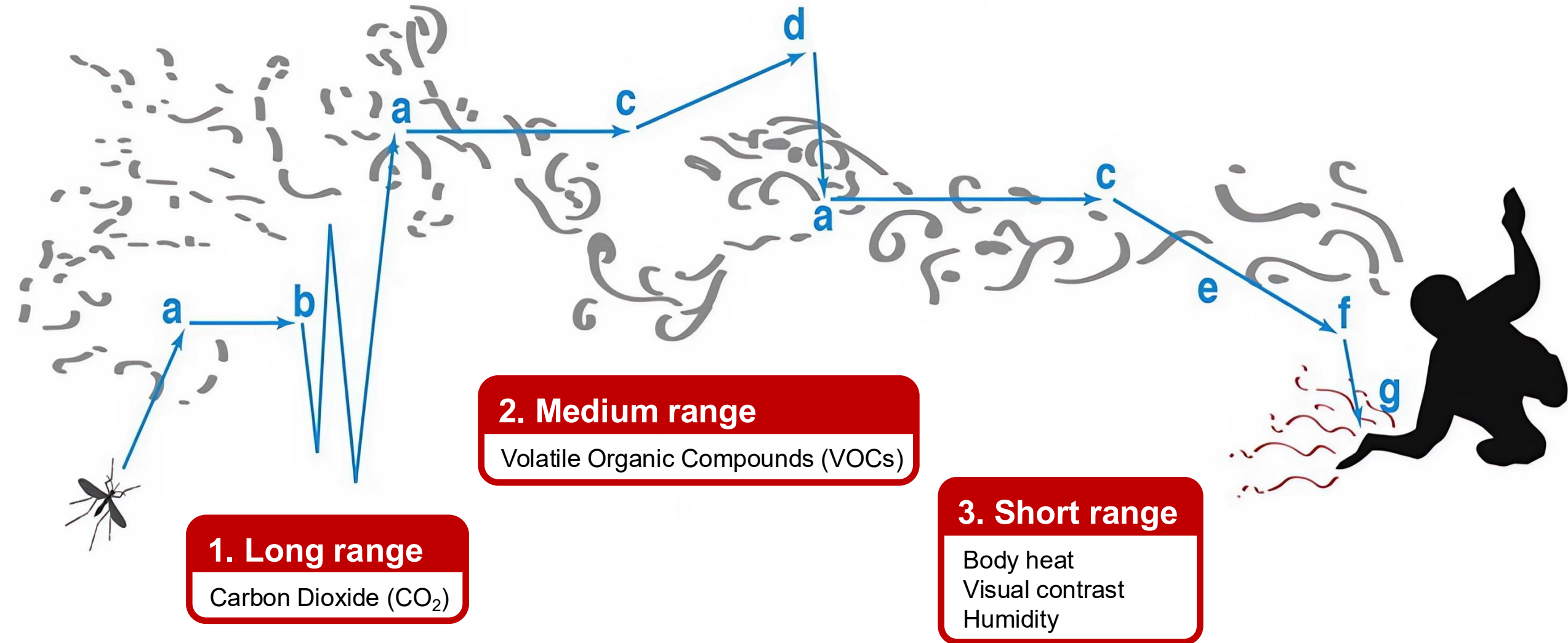
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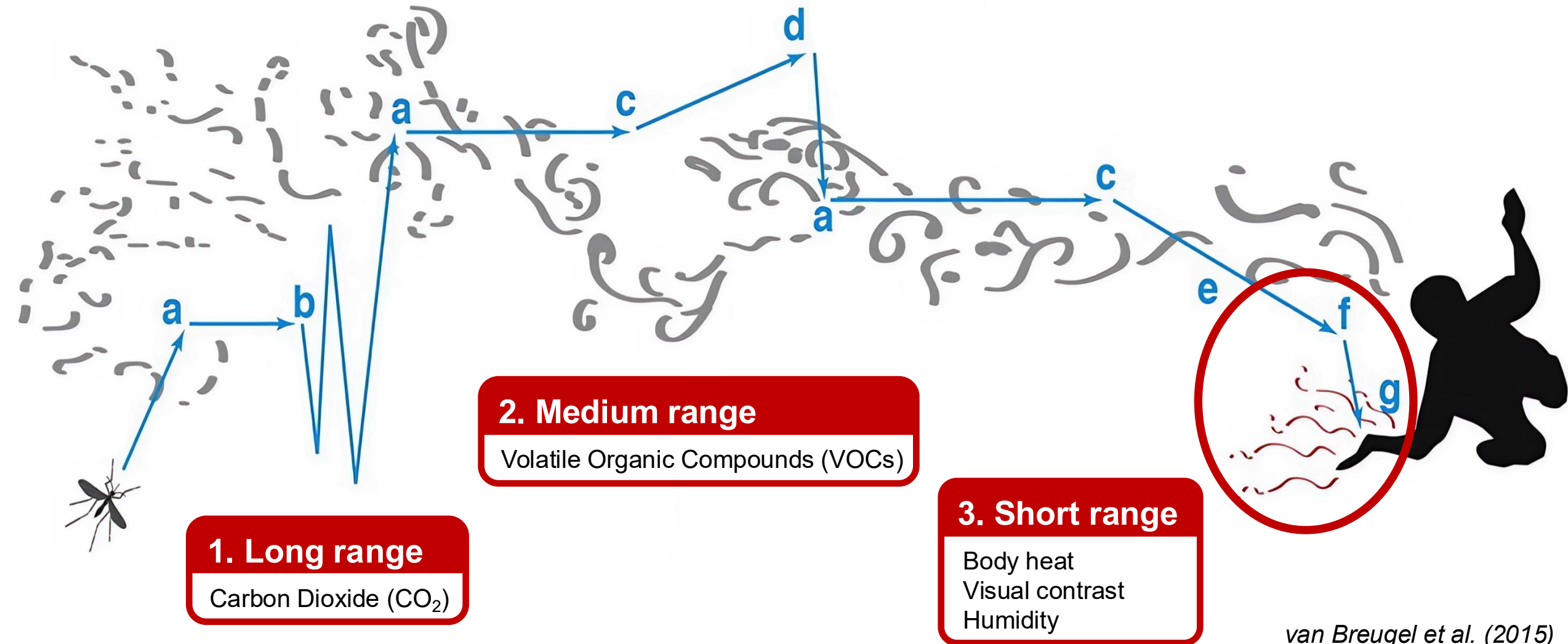
Host location



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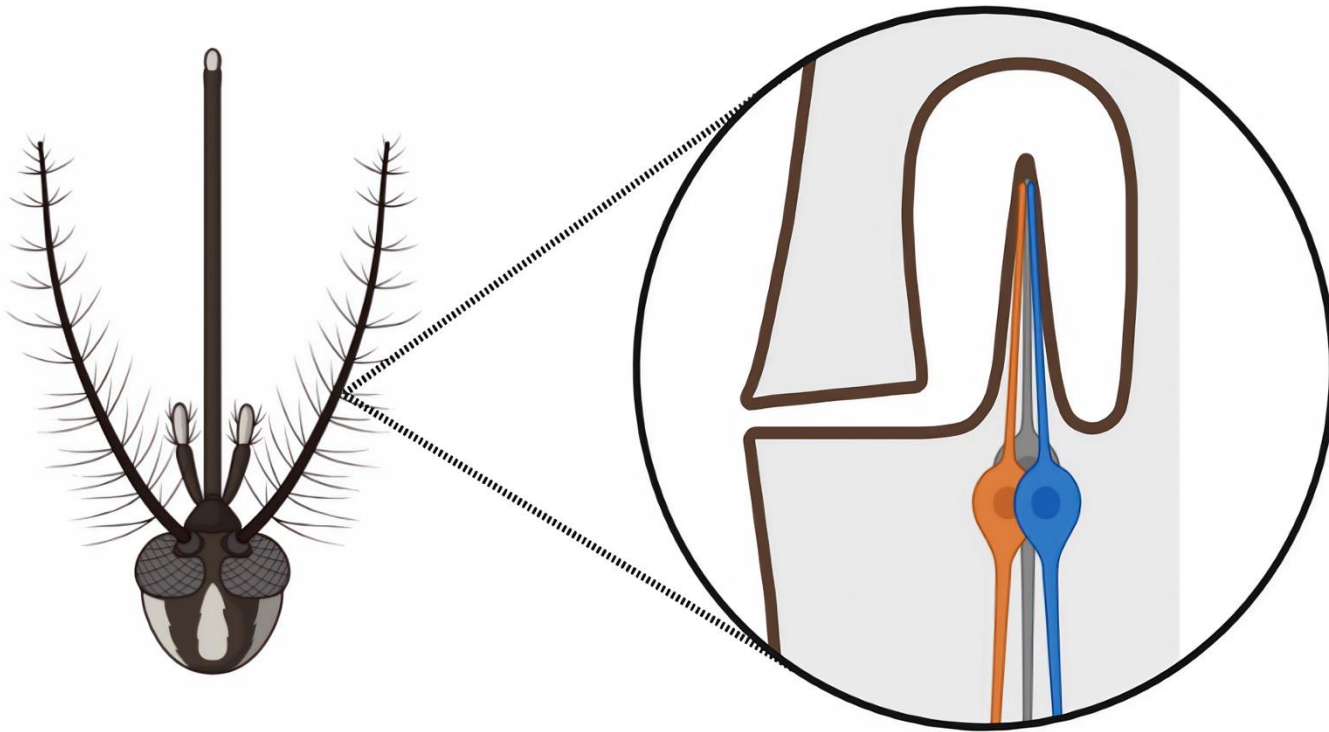
Host location



Humidity sensing in mosquitoes

Humidity sensing in mosquitoes

Hygrosensory sensilla ampullacea



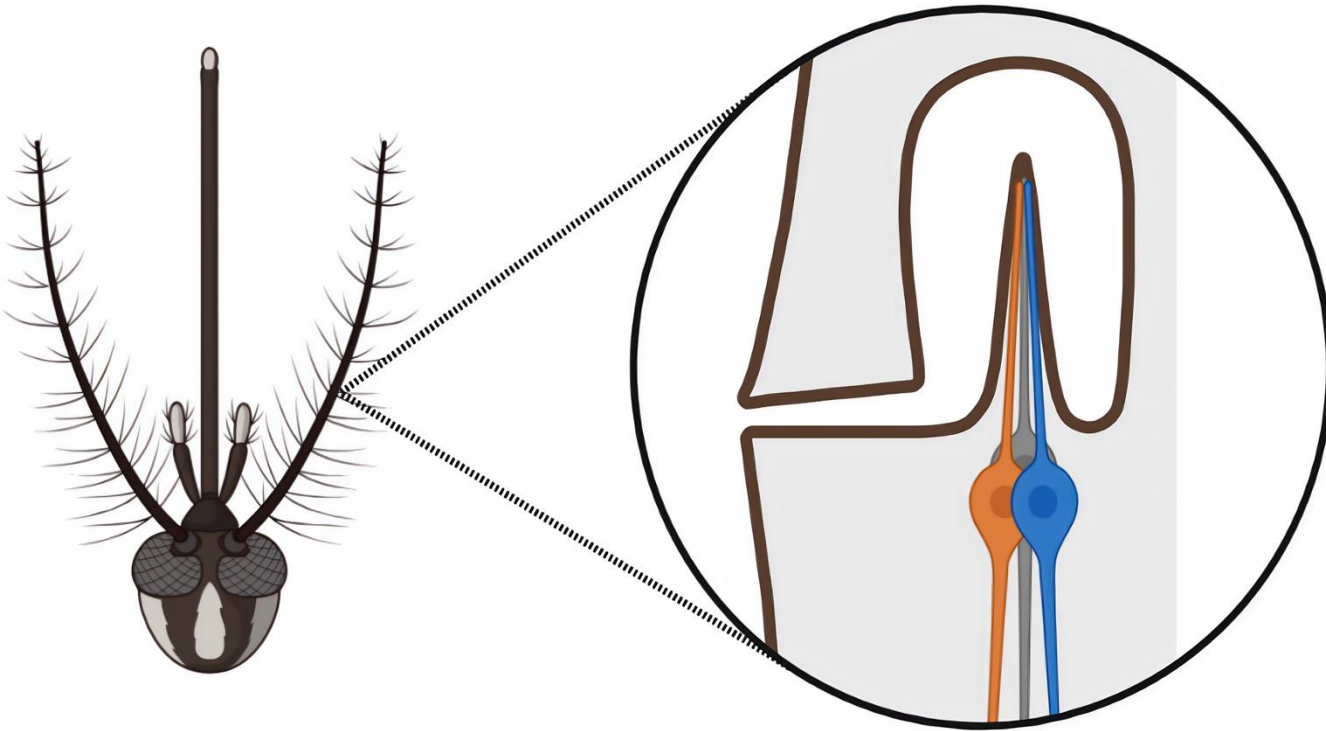
“Dry” cell



“Moist” cell

Humidity sensing in mosquitoes

Hygrosensory sensilla ampullacea



“Dry” cell



“Moist” cell

Ionotropic Receptors

Ir40a

Dry cells only.

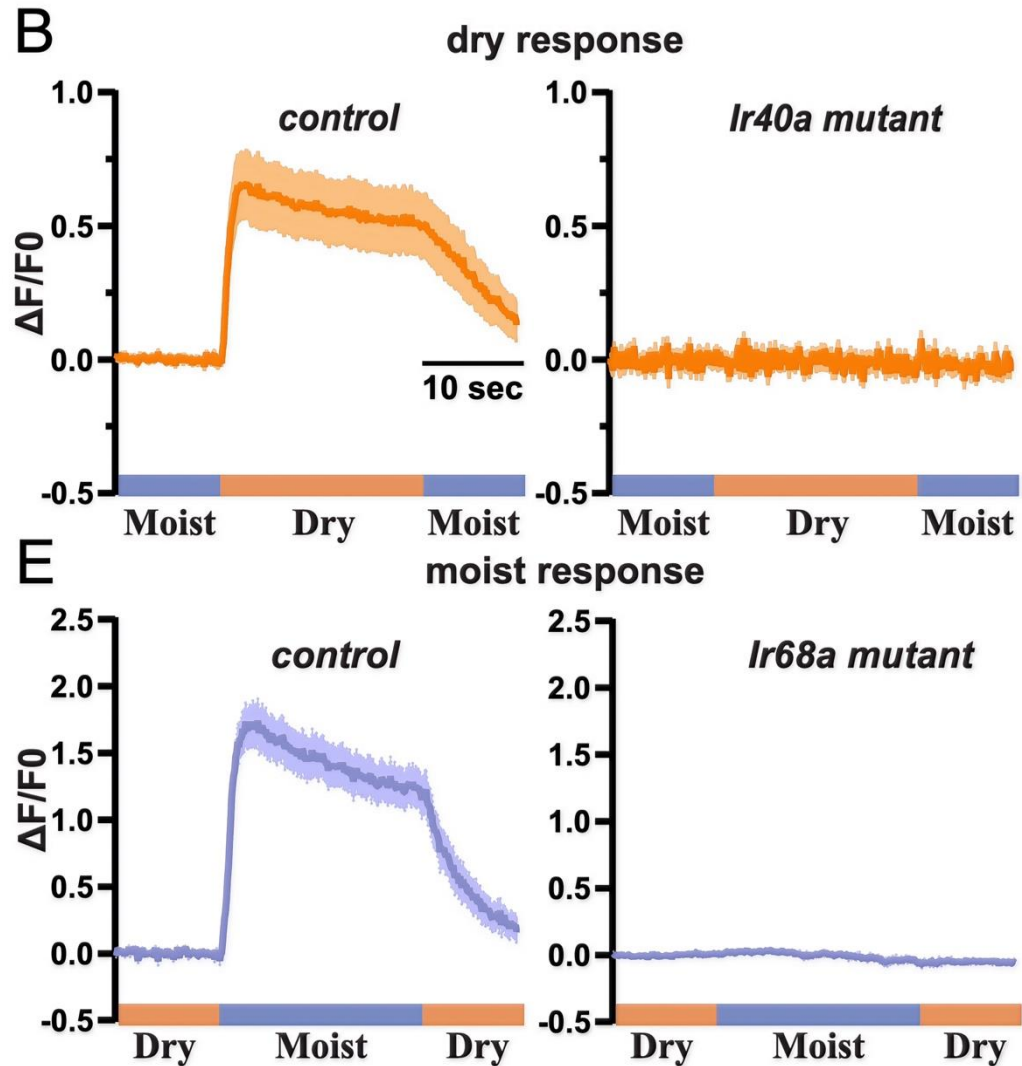
Ir68a

Moist cells only.

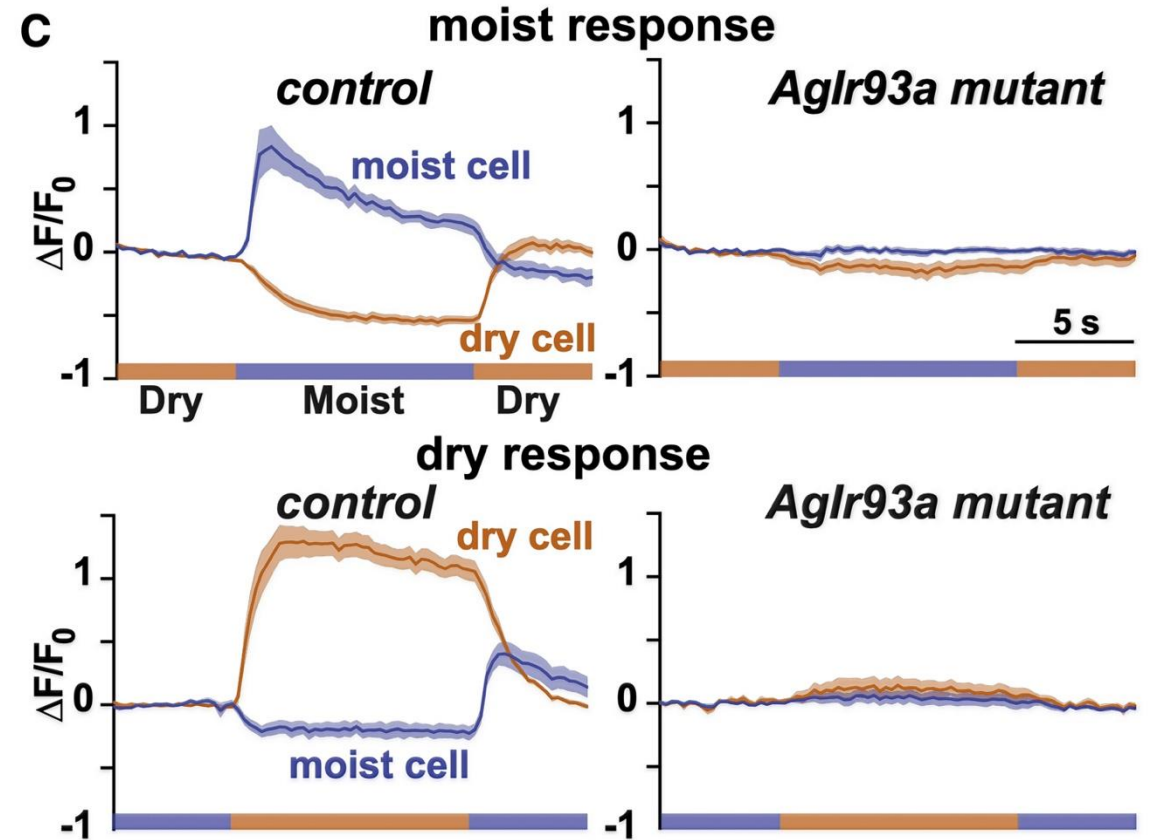
Ir93a

Both **dry** and **moist** cells.

Mosquito hygroreceptors



Tang et al. (2024)



Laursen et al. (2023)

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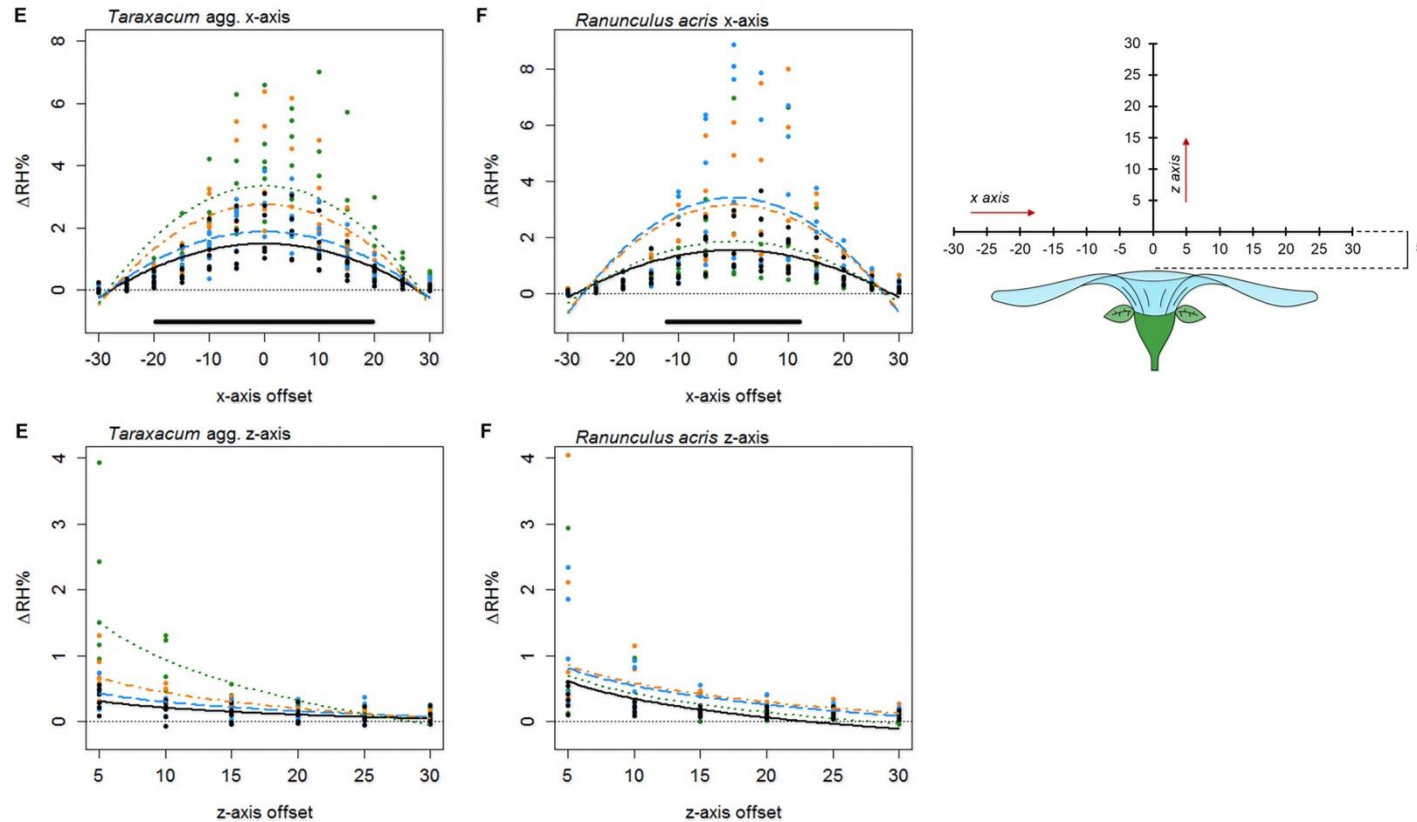
Project plan

Floral humidity

Humidity is elevated around the corolla of many flowering plants as a result of **nectar evaporation** and **petal/stomatal transpiration**.

Floral humidity

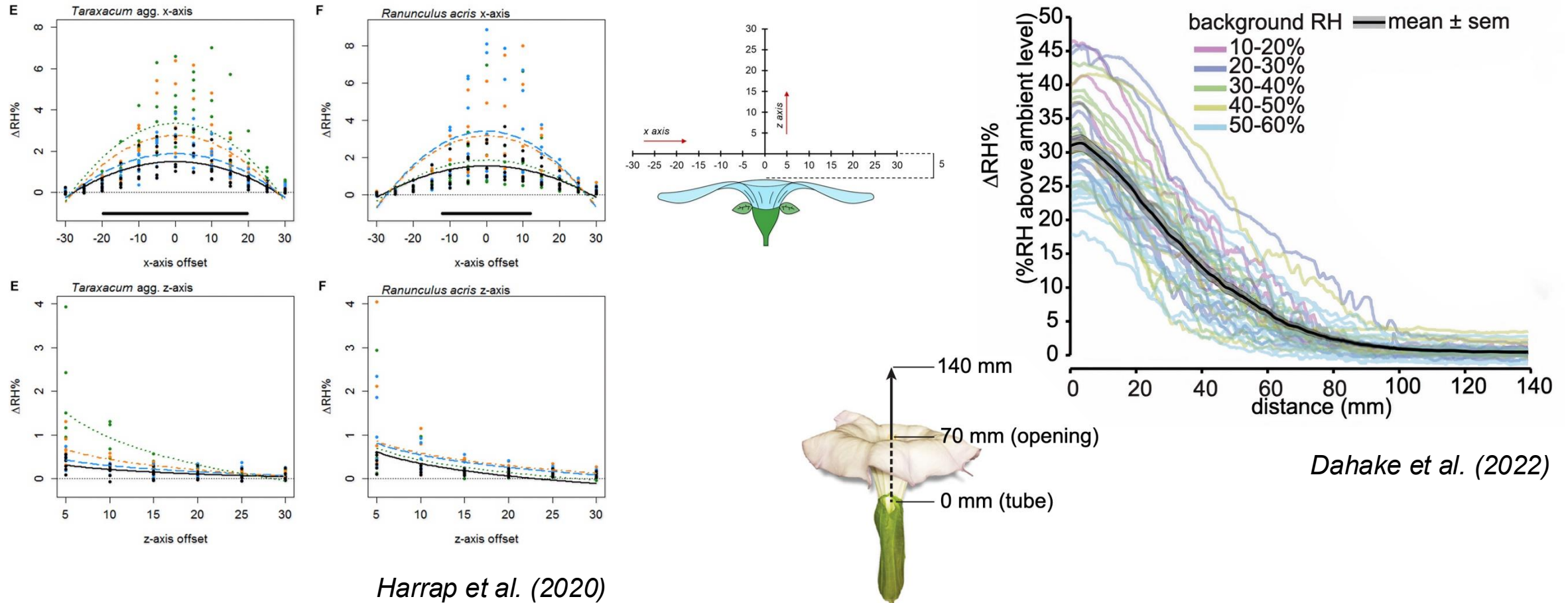
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Harrap et al. (2020)

Floral humidity

Humidity is elevated around the corolla of many flowering plants as a result of **nectar evaporation** and **petal/stomatal transpiration**.



Floral humidity as a foraging cue

Floral humidity as a reliable sensory cue for profitability assessment by nectar-foraging hawkmoths

Martin von Arx^{a,b,1}, Joaquín Goyret^a, Goggy Davidowitz^b, and Robert A. Raguso^a

^aDepartment of Neurobiology and Behavior, Cornell University, Ithaca, NY 14853; and ^bDepartment of Entomology, University of Arizona, Tucson, AZ 85721

Edited by John G. Hildebrand, University of Arizona, Tucson, AZ, and approved May 1, 2012 (received for review January 12, 2012)

Most research on plant–pollinator communication has focused on sensory and behavioral responses to relatively static cues. Floral rewards such as nectar, however, are dynamic, and foraging animals will increase their energetic profit if they can make use of floral cues that more accurately indicate nectar availability. Here we document such a cue—transient humidity gradients—using the night blooming flowers of *Oenothera cespitosa* (Onagraceae). The headspace of newly opened flowers reaches levels of about 4% above ambient relative humidity due to additive evapotranspirational water loss through petals and water-saturated air from the nectar tube. Floral humidity plumes differ from ambient levels only during the first 30 min after anthesis (before nectar is depleted in wild populations), whereas other floral traits (scent, shape, and color) persist for 12–24 h. Manipulative experiments indicated that floral humidity gradients are mechanistically linked to nectar volume and therefore contain information about energy rewards to floral visitors. Behavioral assays with *Hyles lineata* (Sphingidae) and artificial flowers with appropriate humidity gradients suggest that these hawkmoth pollinators distinguish between subtle differences in relative humidity when other floral cues are held constant. Moths consistently approached and probed flowers with elevated humidity over those with ambient humidity levels. Because floral humidity gradients are largely produced by the evaporation of nectar itself, they represent condition-informative cues that facilitate remote sensing of floral profitability by discriminating foragers. In a xeric environment, this level of honest communication should be adaptive when plant reproductive success is pollinator limited, due to intense competition for the attention of a specialized pollinator.

pollination biology | honest signaling | floral display

Plant–pollinator relationships—like all mutualisms—strike a tenuous balance between the conflicting interests of foraging animals and flowering plants (1, 2). For flowering plants, pollinator attraction through floral density, form, color, pattern, and odor is necessary but insufficient to ensure effective pollination. Ideally, pollinators should transfer pollen between conspecific plants in ways that result in favorable levels of outcrossing (3) and reduced pollen loss (4). Flowers commonly manipulate pollinator movement by varying quality or quantity of floral nectar. The offer of variable amounts of nectar (5, 6), toxic nectar (7, 8), or an empty promise of nectar elicit similar results: pollinators leave plants sooner, reducing for example geitonogamous inbreeding (if self-compatible) or pollen discounting (if self-incompatible).

The availability of floral rewards often coincides with floral display and pollinator activity on a gross temporal scale (i.e., hours) (9, 10). Nevertheless, most pollinators encounter reward-

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RESEARCH ARTICLE

Bumblebees can detect floral humidity

Michael J. M. Harrap^{1,2,*}, Natalie Hempel de Ibarra², Henry D. Knowles^{1,3}, Heather M. Whitney¹ and Sean A. Rands¹

ABSTRACT

Floral humidity, a region of elevated humidity in the headspace of the flower, occurs in many plant species and may add to their multimodal floral displays. So far, the ability to detect and respond to floral humidity cues has been only established for hawkmoths when they locate and extract nectar while hovering in front of some moth-pollinated flowers. To test whether floral humidity can be used by other more widespread generalist pollinators, we designed artificial flowers that presented biologically relevant levels of humidity similar to those shown by flowering plants. Bumblebees showed a spontaneous preference for flowers that produced higher floral humidity. Furthermore, learning experiments showed that bumblebees are able to use differences in floral humidity to distinguish between rewarding and non-rewarding flowers. Our results indicate that bumblebees are sensitive to different levels of floral humidity. In this way floral humidity can add to the information provided by flowers and could impact pollinator behaviour more significantly than previously thought.

KEY WORDS: Behaviour, Floral Display, Multimodal cues, Learning, Bumblebees, Angiosperms, Flower, Pollination

INTRODUCTION

Floral humidity, an area of elevated humidity within the headspace of the flower, has been demonstrated to occur in a number of flower species (Corbet et al., 1979; Nordström et al., 2017; von Arx et al., 2012). Floral humidity is created by a combination of nectar evaporation and floral transpiration (Azad et al., 2007; Corbet et al., 1979; Harrap et al., 2020a; von Arx et al., 2012), although the contribution of these two influences may vary between species. Transects of the flower headspace of 42 species found 30 (71%) produce floral humidity of an intensity greater than would be

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Article

<https://doi.org/10.1038/s41467-022-35353-8>

A signal-like role for floral humidity in a nocturnal pollination system

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 Check for updates

Ajinkya Dahake^{1✉}, Piyush Jain², Caleb C. Vogt¹, William Kandalaft¹, Abraham D. Stroock³ & Robert A. Raguso¹

Previous studies have considered floral humidity to be an inadvertent consequence of nectar evaporation, which could be exploited as a cue by nectar-seeking pollinators. By contrast, our interdisciplinary study of a night-blooming flower, *Datura wrightii*, and its hawkmoth pollinator, *Manduca sexta*, reveals that floral relative humidity acts as a mutually beneficial signal in this system. The distinction between cue- and signal-based functions is illustrated by three experimental findings. First, floral humidity gradients in *Datura* are nearly ten-fold greater than those reported for other species, and result from

von Arx et al. (2012)

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Despite numerous behavioural studies, a mechanism hasn't yet been proposed.

von Arx et al. (2012)

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So why mosquitoes?

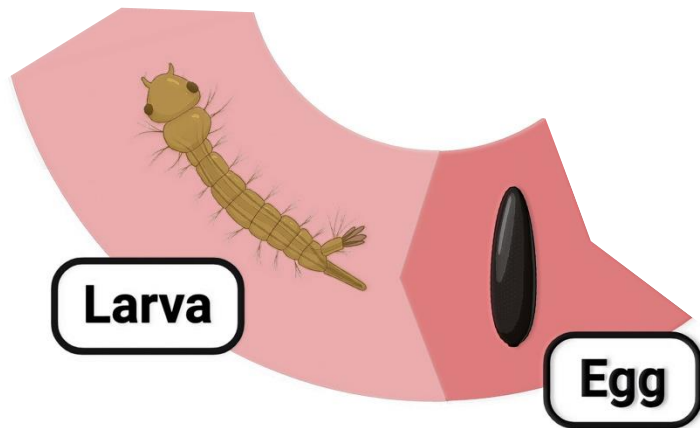
Mosquito life cycle

So why mosquitoes?



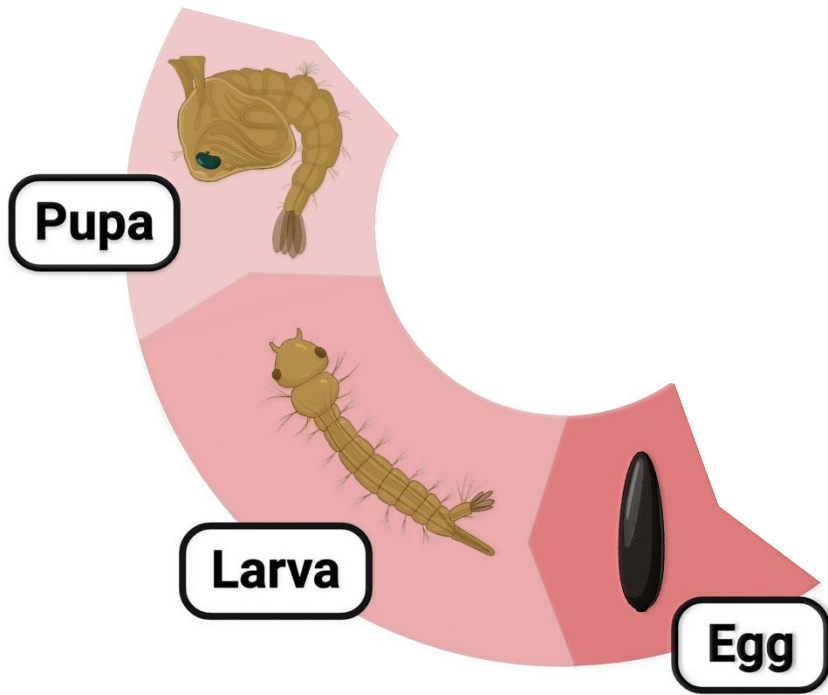
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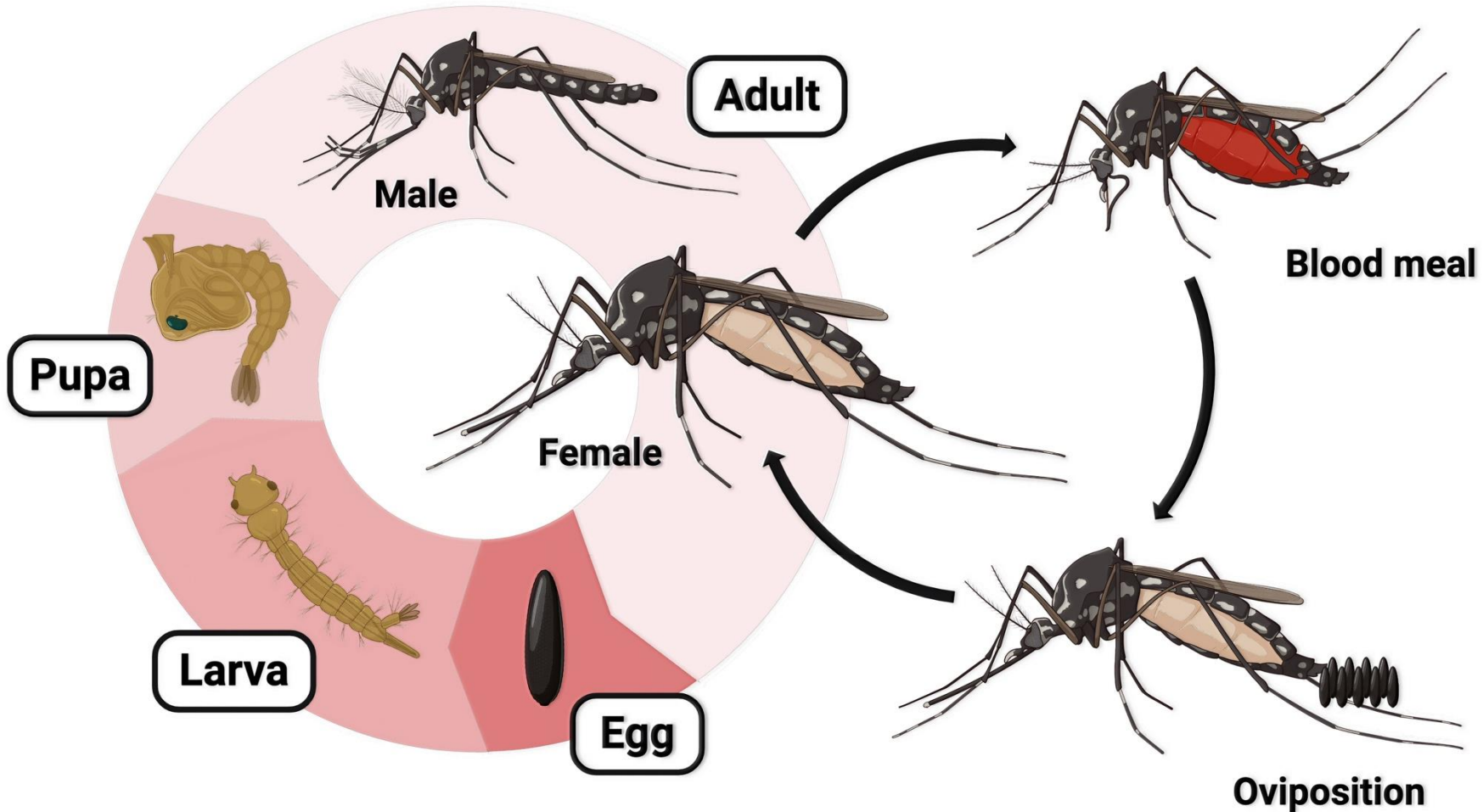
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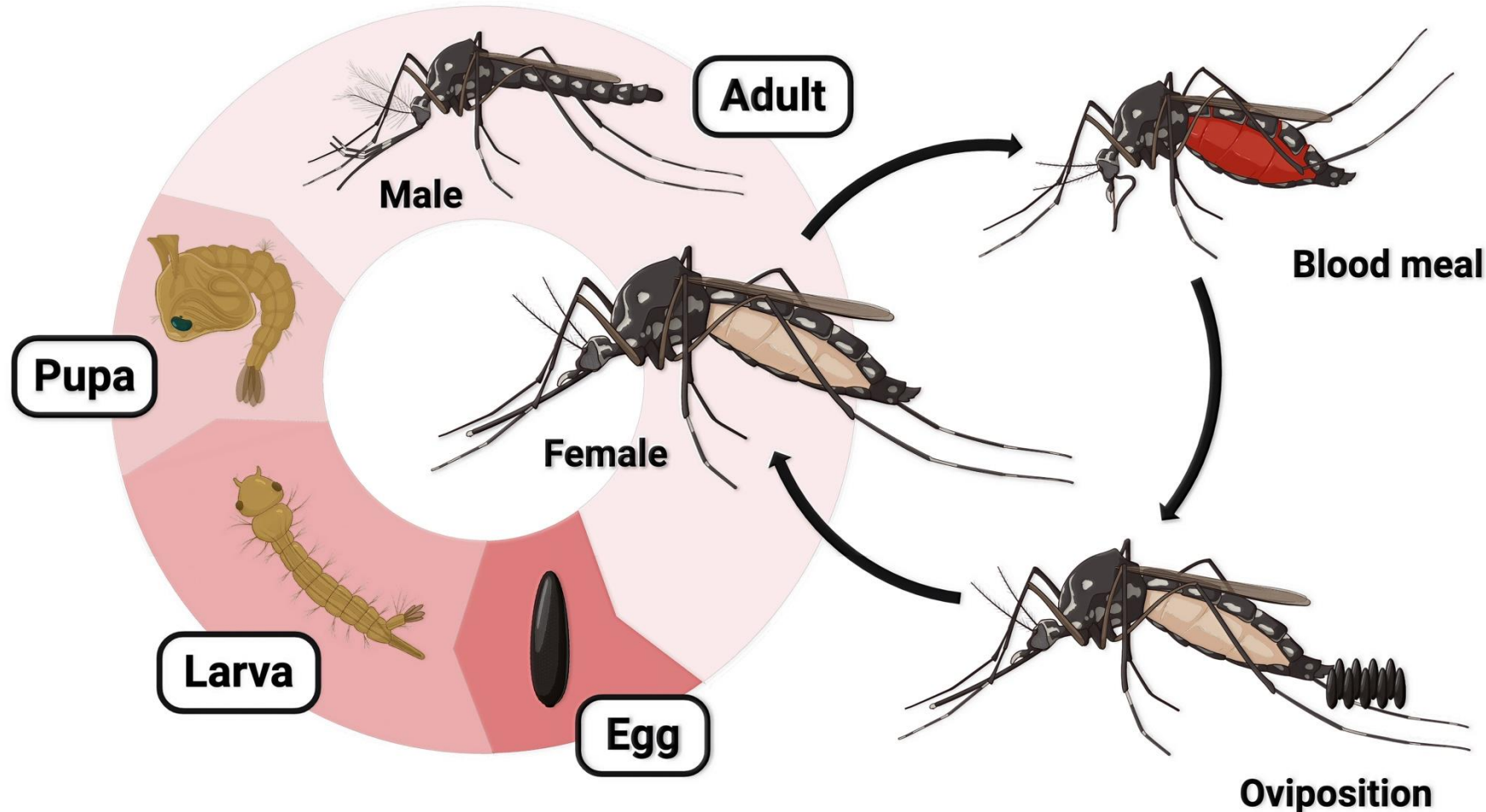
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Hygroreceptor involvement in **gonotrophic cycle** well characterised...

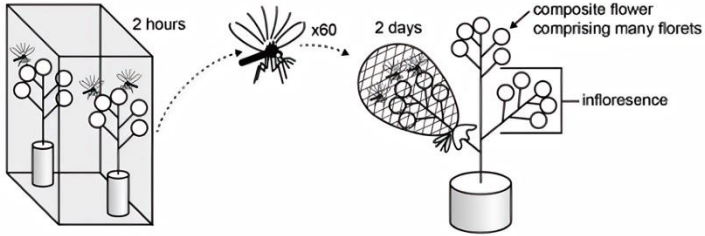
but their role in **nectar-feeding** has is yet to be explored.

Mosquitoes as pollinators

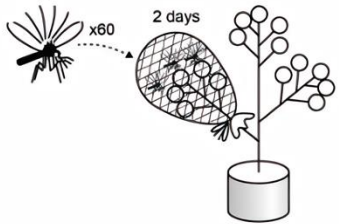
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Pollination experiment #1

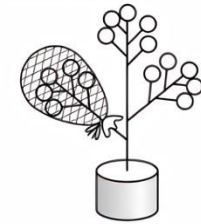
Cross-pollination



Self-pollination

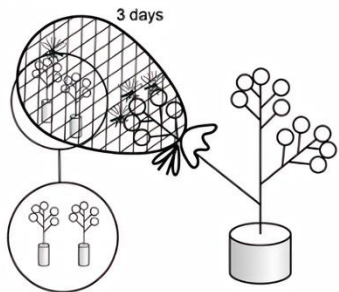


Blank control

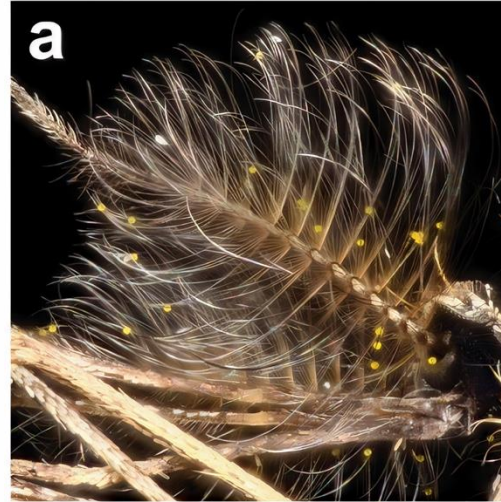
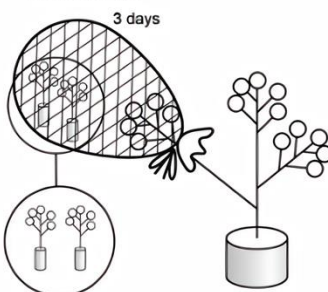


Pollination experiment #2

Cross-pollination



Blank control

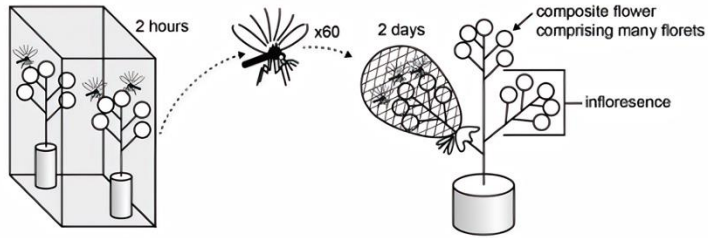


Peach and Gries (2016)

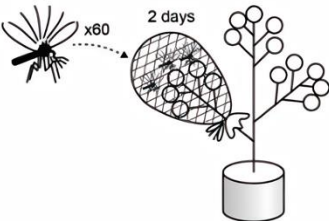
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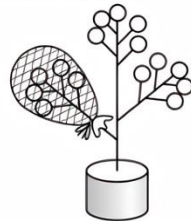
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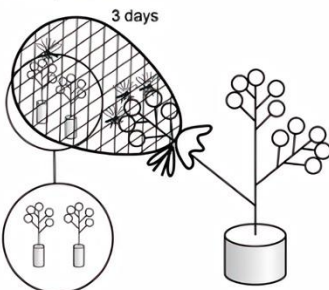


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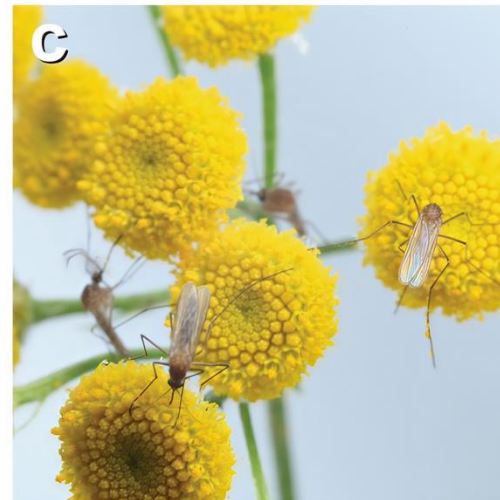
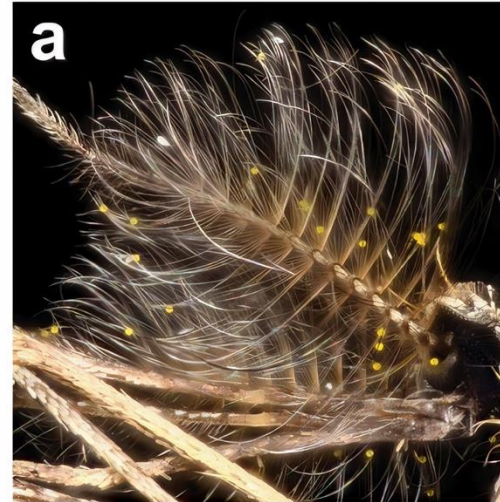
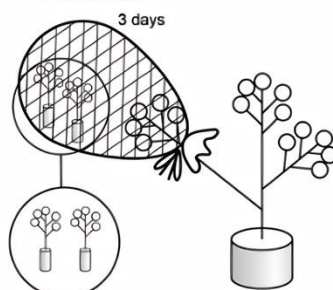


Pollination experiment #2

Cross-pollination



Blank control



The olfactory basis of orchid pollination by mosquitoes

Chloé Lahondère^{a,1}, Clément Vinauer^{a,1}, Ryo P. Okubo^a, Gabriella H. Wolff^a, Jeremy K. Chan^a, Omar S. Akbari^{a,b}, and Jeffrey A. Riffell^{a,2}

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Edited by Walter S. Leal, University of California, Davis, CA, and accepted by Editorial Board Member John R. Carlson November 22, 2019 (received for review June 20, 2019)

Mosquitoes are important vectors of disease and require sources of carbohydrates for reproduction and survival. Unlike host-related behaviors of mosquitoes, comparatively less is understood about the mechanisms involved in nectar-feeding decisions, or how this sensory information is processed in the mosquito brain. Here we show that *Aedes* spp. mosquitoes, including *Aedes aegypti*, are effective pollinators of the *Platanthera obtusata* orchid, and demonstrate this mutualism is mediated by the orchid's scent and the balance of excitation and inhibition in the mosquito's antennal lobe (AL). The *P. obtusata* orchid emits an attractive, nonanal-rich scent, whereas related *Platanthera* species—not visited by mosquitoes—emit scents dominated by lilac aldehyde. Calcium imaging experiments in the mosquito AL revealed that nonanal and lilac aldehyde each respectively activate the LC2 and AM2 glomeruli, and remarkably, the AM2 glomerulus is also sensitive to N,N-diethyl-methyl-tolamide (DEET), a mosquito repellent. Lateral inhibition between these 2 glomeruli reflects the level of attraction to the orchid scents. Whereas the enriched nonanal scent of *P. obtusata* activates the LC2 and suppresses AM2, the high level of lilac aldehyde in the other orchid scents inverts this pattern of glomerular activity, and behavioral attraction is lost. These results demonstrate the ecological importance of mosquitoes beyond operating as disease vectors and open the door toward understanding the neural basis of mosquito nectar-seeking behaviors.

the flowers, and the importance of mosquito visitation for orchid pollination, are unknown.

In this article, we examine the neural and behavioral processes mediating mosquito floral preference. We present findings from 1) pollination studies in *P. obtusata* by *Aedes* mosquitoes, 2) analyses of floral scent compounds that attract diverse mosquito species, and 3) antennal and antennal lobe (AL) recordings showing how these floral scents and compounds are represented in the mosquito brain (SI Appendix, Fig. S1). Using this integrative approach, we demonstrate that *Aedes* discrimination of *Platanthera* orchids is mediated by the balance of excitation and inhibition in the mosquito antennal lobe.

Results
To understand the importance of various pollinators, including mosquitoes, on *P. obtusata*, we first conducted pollinator observation and exclusion experiments in northern Washington State where *Platanthera* orchids and mosquitoes are abundant. Using a combination of video recordings and focal observations by trained participants, more than 581 *P. obtusata* flowers were observed for a total of 47 h, with 57 floral feeding events by mosquitoes. During our observations, flowers were almost solely visited by various mosquito species (both sexes) that mainly belonged to the *Aedes* group (Fig. 1A and B and SI Appendix, Table S2), with the only

Platanthera | mosquitoes | *Aedes aegypti* | olfaction | nectar

Significance

Nectar feeding by mosquitoes is important for survival and reproduction, and hence disease transmission. However, we know little about the sensory mechanisms that mediate mosquito attraction to sources of nectar, like those of flowers, or how this information is processed in the mosquito brain. Using a unique mutualism between *Aedes* mosquitoes and *Platanthera obtusata* orchids, we reveal that the orchid's scent mediates this mutualism. Furthermore, lateral inhibition in the mosquito's antennal (olfactory) lobe—via the neurotransmitter GABA—is critical for the representation of the scent. These results have implications for understanding the olfactory basis of mosquito nectar-seeking behaviors.

Author contributions: C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. designed research; C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. performed research; O.S.A. contributed new reagents/analytic tools; C.L., C.V., R.P.O., G.H.W., J.K.C., and J.A.R. analyzed data; and C.L., C.V., R.P.O., G.H.W., J.K.C., O.S.A., and J.A.R. wrote the paper.

Competing interest statement: A provisional patent on the odor that mimics the orchid scent was recently filed (SI Appendix, Table S2).

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Data deposition: Data is available on Mendeley Data (<https://data.mendeley.com/datasets/brn42q9f1>).

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Lahondère et al. (2020)

Peach and Gries (2016)

Overview

Mosquitoes as vectors for disease

Mosquito humidity sensing

Floral humidity and insect pollinators

Mosquitoes as pollinator models

Project plan

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1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.

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Expanded study

1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.
3. Normalise expression levels between datasets.

Project plan

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To determine whether hygropceptors are involved in mosquito nectar-foraging.

Pilot study

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3. Identify protein families using OrthoFinder.
4. Compare transcriptomic expression of key hygropceptors between sexes over their life cycle using DESeq2 in R.

Expanded study

1. Expand transcriptomic dataset with additional mosquito species.
2. Pre-process data, assemble transcriptomes, quantify expression and identify protein families.
3. Normalise expression levels between datasets.
4. Compare transcriptomic expression of hygropceptors across mosquito species as well as that of CO₂ and VOC receptors.

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- Floral humidity acts as a foraging cue for insect pollinators but a mechanism for this hasn't yet been established.
- Mosquitoes offer an exciting opportunity to study this due to their complex life cycle which varies between sexes.

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Any questions?